

NUST

NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY

Name: M Abdullah

Department/Batch:

IGIS-2024

CMS Id: 501223

Date:1 Dec,2025

Course: DMIP

Submitted to: Sir Ejaz

Supervised Classification Lab Report

Title: Land Use/Land Cover Classification Using Supervised Classification Method

1. Introduction

Supervised classification is a widely used technique in remote sensing for categorizing pixels in satellite imagery into distinct land cover classes. This method relies on user-defined training samples to create spectral signatures that represent different land cover types. The software then uses these signatures to classify all pixels in the image based on their spectral characteristics.

This laboratory exercise demonstrates the application of supervised classification using the Maximum Likelihood algorithm on a Landsat ETM+ satellite image to identify and map five major land use categories in the study area.

2. Objectives

The primary objectives of this laboratory exercise were:

- To generate supervised signatures using Area of Interest (AOI) and Region Grow tools
- To create training samples for different land cover classes
- To use signature files as input for Maximum Likelihood classification
- To classify satellite imagery into five distinct land use types
- To evaluate the accuracy and effectiveness of the supervised classification method

3. Study Area

The study area consists of a subset of a Landsat Enhanced Thematic Mapper Plus (ETM+) image. Landsat ETM+ is a multispectral sensor that captures data across seven spectral bands, providing valuable information for land cover analysis. The image contains diverse land cover types including water bodies, urban infrastructure, residential areas, forested regions, and vegetation.

4. Methodology

4.1 Software and Data

- **Software:** ERDAS Imagine
- **Image Data:** Landsat ETM+ subset (study area final.img)
- **Classification Method:** Supervised Classification
- **Classification Algorithm:** Maximum Likelihood

4.2 Classification Scheme

The following classification scheme was developed for this study:

Land Cover Type	Class Code
Water	1
Roads	2
Residential	3
Forest	4
Vegetation	5

4.3 Procedure

Step 1: Image Display

- Opened the Landsat ETM+ image in ERDAS Imagine Viewer
- Displayed the image using "Fit to Frame" option for optimal visualization

Step 2: Signature Editor Initialization

- Accessed the Signature Editor through Raster → Supervise → Signature Editor
- Prepared the interface for creating training samples

Step 3: Training Sample Collection

- Used the polygon drawing tool to delineate representative samples for each land cover class
- Selected multiple training areas for each class across the entire image to ensure spectral variability was captured
- For each class, the following process was followed:
    - Drew polygons around homogeneous areas representing the target class
    - Added signatures to the Signature Editor
    - Selected all signatures for the same class
    - Merged multiple signatures into a single representative signature
    - Renamed the merged signature according to the land cover type
    - Deleted individual signatures to maintain organized signature file

Step 4: Signature File Creation

- Repeated the training sample collection process for all five land cover classes
- Ensured adequate spatial distribution of training samples
- Saved the signature file with an appropriate name for future reference

Step 5: Classification Execution

- Accessed Supervised Classification through Raster → Supervised Classification
- Specified input image and output file locations
- Selected Maximum Likelihood as the parametric classification rule
- Deselected all signatures in the Signature Editor to ensure all classes were included

- Executed the classification with default parameters

Step 6: Results Visualization

- Displayed the classified image alongside the original image using Geospatial Light Table
- Compared the classification results with the original imagery for visual accuracy assessment

5. Results and Discussion

5.1 Classification Output

The supervised classification successfully categorized the study area into five distinct land use/land cover classes. The Maximum Likelihood classifier assigned each pixel to the class with the highest probability based on the training signatures.



5.2 Visual Assessment

A visual comparison between the original Landsat image and the classified output revealed:

- **Water bodies** were clearly delineated and showed high classification accuracy
- **Roads** were identified, though some narrow roads may have been missed due to pixel resolution
- **Residential areas** were distinguished from other urban features
- **Forest** areas showed distinct classification from other vegetation types
- **Vegetation** (non-forest) was successfully separated from forested regions

5.3 Advantages of Supervised Classification

- User control over class definitions and training sample selection
- Incorporation of prior knowledge about the study area
- Ability to target specific land cover classes of interest
- Generally higher accuracy when adequate training data is provided

5.4 Limitations

- Time-consuming training sample collection process
- Requires prior knowledge or ground truth data
- Classification accuracy depends on the quality of training samples
- Mixed pixels at class boundaries may be misclassified

---

6. Conclusion

This laboratory exercise successfully demonstrated the application of supervised classification for land use/land cover mapping. The Maximum Likelihood algorithm effectively classified the Landsat ETM+ image into five distinct categories based on spectral signatures derived from training samples.

The supervised classification approach proved effective for distinguishing between different land cover types in the study area. The method's reliance on user-defined training samples allowed for targeted classification of specific land cover classes relevant to the study objectives.

Future improvements could include:

- Collection of ground truth data for accuracy assessment
- Expansion of training samples to improve classification reliability
- Post-classification refinement to reduce classification errors
- Comparison with unsupervised classification results